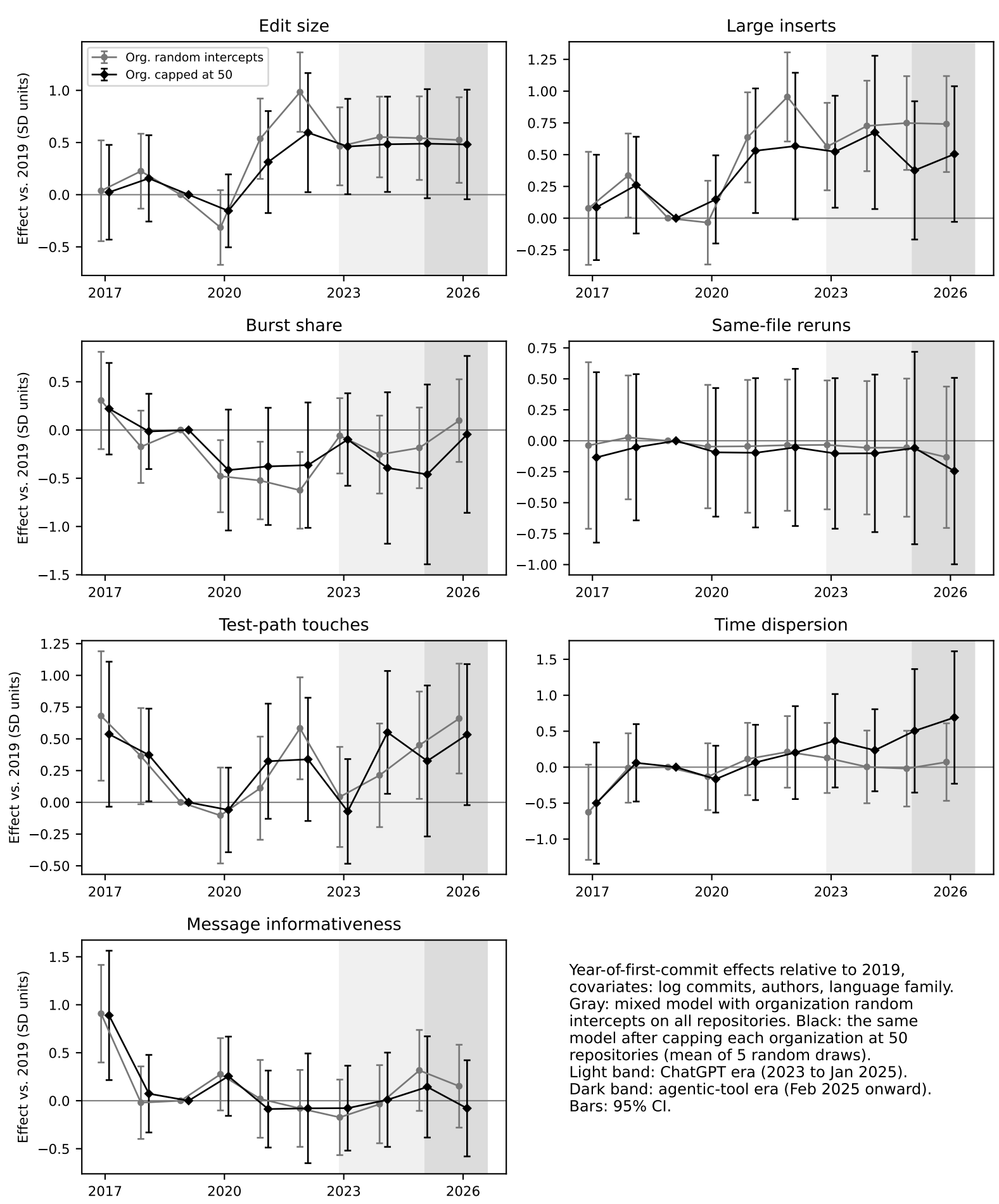

Year-of-first-commit effects relative to 2019, covariates: log commits, authors, language family. Gray: mixed model with organization random intercepts on all repositories. Black: the same model after capping each organization at 50 repositories (mean of 5 random draws). Light band: ChatGPT era (2023 to Jan 2025). Dark band: agentic-tool era (Feb 2025 onward). Bars: 95% CI.